User Interface Design

Designing User Interface

- An effective communication medium between a human and a computer
- Identification of interface objects and actions
- Creation of a screen layout
- Study of people and how they relate to technology by answering questions like:
  - Who is the user?
  - How does the user learn to interact with the system?
  - How does the user interpret info produced by the system?
  - What will the user expect of the system?

Golden Rules

- Place the User in Control
- Reduce User's Memory Load
- Make the Interface Consistent

Place the user in control

- System should react to user needs
- System should help the user complete tasks
- User should not feel that the system is controlling the user

Place the user in control

- Design Principles:
  - Define interactions s.t. a user is not forced into unnecessary/undesired actions/modes
  - Provide flexible interaction
  - User should not feel that the system is controlling the user
  - Allow interruptible and undoable user interactions
  - Streamline interactions based on skill level, allow interactions to be customized
  - Hide technical internals from casual user
  - Provide mechanism for direct interaction with objects on screen

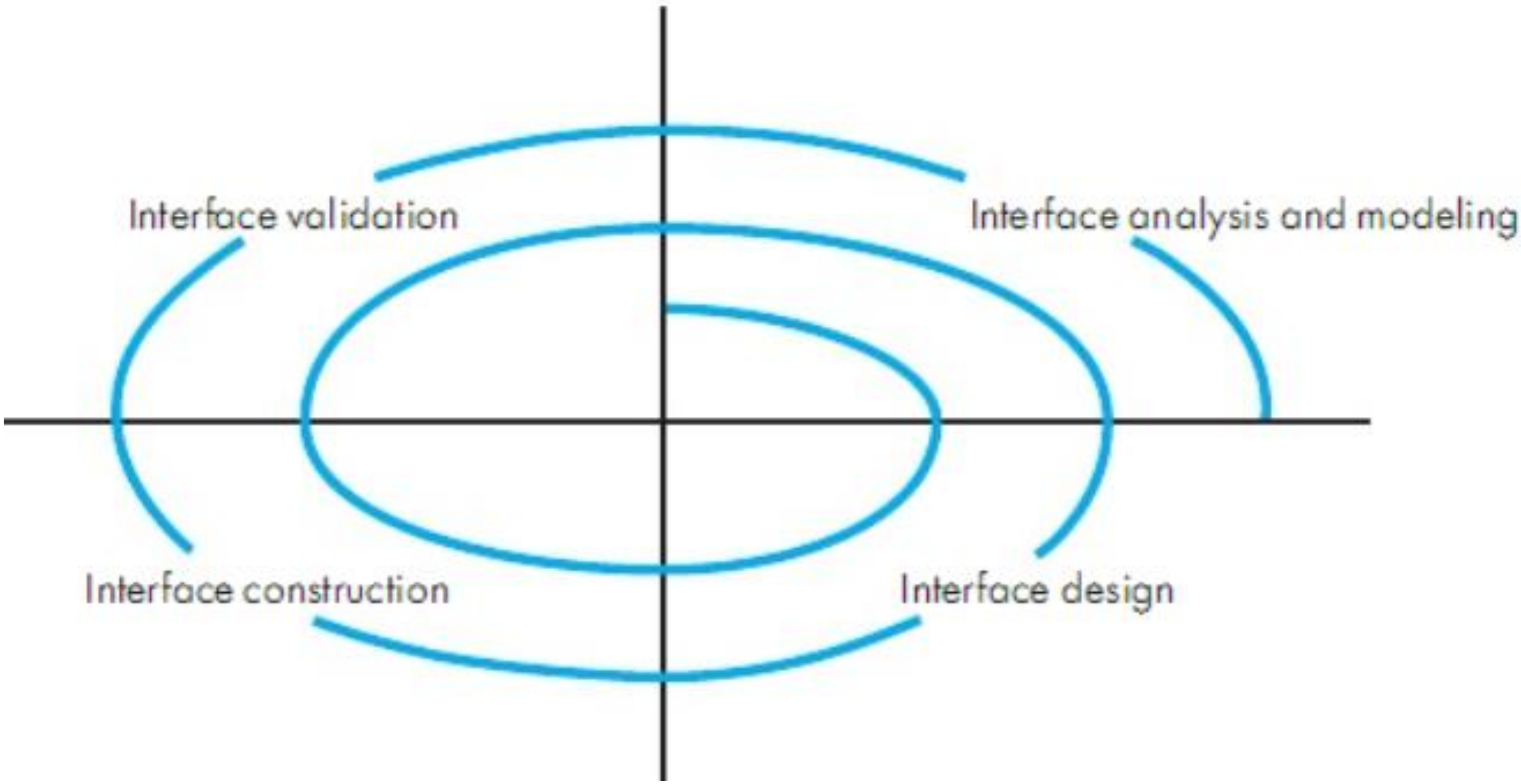
Reduce the user's memory load

- The more a user has to remember, the more error-prone the interaction
- Design principles:
  - Reduce demand on short term memory
  - Establish meaningful defaults
  - Define shortcuts that are intuitive
  - Disclose information in a progressive manner

Make the interface consistent

- Maintain design rules for all screens
- Design principles:
  - Allow user to put current task into a meaningful context
  - Maintain consistency across a complete product line
  - Avoid violating de facto standards

UID Process


Interface validation
Interface analysis and modeling
Interface construction
Interface design

UID Process

- Interface Analysis
  - User Model
    - Profile of end user
      - Age, gender, education, physical abilities, cultural/ethnic background etc.
    - Novice, Knowledgeable, Knowledgeable frequent users
  - Design Model
  - Mental Model
    - User’s perception of the system
  - Implementation Model

UID Process

- Interface Design
  - Define interface objects and actions
- Interface Construction
  - Creation of prototypes to evaluate usage scenarios
- Interface Validation
  - Ability to implement every user task correctly
  - Degree to which interface is easy to use
  - User’s acceptance to user interface as useful

Interface Analysis

- User Analysis
  - Trained professionals? Technicians? Clerks? Etc.
  - Average level of formal education?
  - User capability to learn from written material/training?
  - Expert typists? Do not like keyboard?
  - Age range?
  - Users represented predominantly by one gender?
  - Routine of work? Regular? Overtime?
  - Frequency of usage?
  - Primary spoken language of users?
  - Subject matter experts?
  - Desire to know underlying technology

Interface Analysis

- Task Analysis
  - Work performed by user in particular circumstances
  - Tasks/subtasks performed by the user during the work being performed
  - Problem domain objects manipulated by users
  - Sequence of work tasks? Flow of actions? The workflow?
  - Hierarchy of tasks

  - Techniques used to support task analysis
    - Usecases
    - Task Elaboration
    - Object Elaboration
    - Workflow Analysis(Swimlane Diagram)

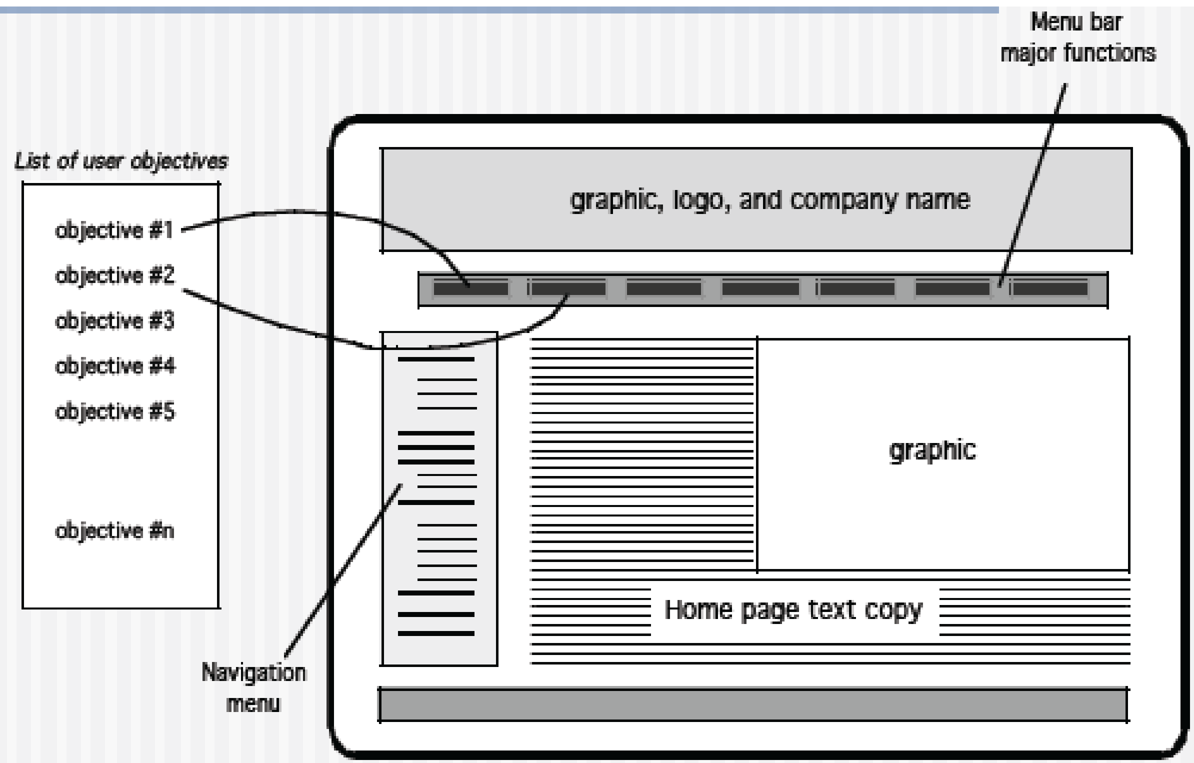
Interface Analysis

- Analysis and display content
  - Different types of data assigned to consistent locations on screen?
  - Customizable location of content on screen
  - Partitioning of large objects (e.g. reports) for better understanding
  - Use of colors to improve understanding
  - Content of error messages?

Interface Design

- Define interface objects and operations
- Model system behavior on events that change state of the interface
- Indicate user interpretation of the state of the system

Mapping User Objectives



Design Issues

- Response time
  - Length
  - Variability
    - Interaction rhythm
- Help
  - Help without leaving the interface
- Error information handling
- Command labeling
  - Power users?

Task	Response time (seconds)
1	2
2	1
3	3
4	4
5	3
Mean	
Standard deviation	

Task	Response time (seconds)
1	1.5
2	1
3	2
4	1.25
5	2.25
Mean	
Standard deviation	

Other Design Considerations
Designing User Interfaces

- Must consider several issues:
  - identifying the humans who will interact with the system
  - defining scenarios for each way that the system can perform a task
  - designing a hierarchy of user commands
  - refining the sequence of user interactions with the system
  - designing relevant classes in the hierarchy to implement the user-interface design
  - decisions
  - integrating the user-interface classes into the overall system class hierarchy

UI Trends

- Minimalism
- Skeumorphism
- Laser Focus
- Context Sensitive Navigation
- Collapsed Content
- Content Chunking
- Long Pages
- For details see http://www.authorstream.com/Presentation/umr17-1645231-pressman-ch-12-user-interface-design/

Design Evaluation

- Evolutionary…

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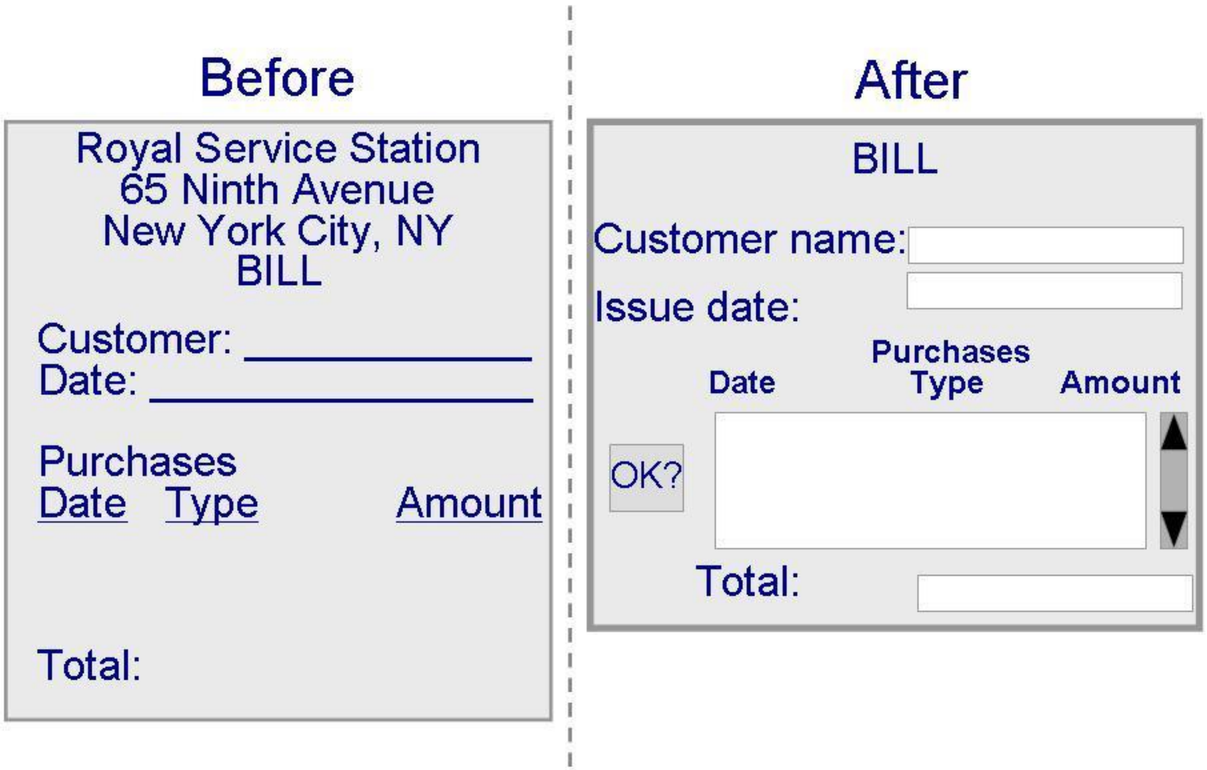
Other Design Considerations
Designing User Interfaces (continued)



References

- UCF slides for SE course (a few slides have been reused)
- SE book by Pressman
- SE book by Pfleeger
- SE book by Ian Sommerville